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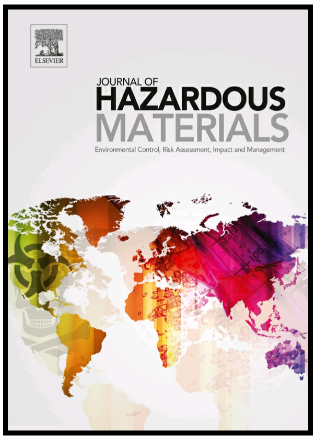

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Profiling of Emerging Contaminants and Antibiotic Resistance in Sewage Treatment Plants: An Indian Perspective

Priyam Saxena[a#], Isha Hiwrale[a#], Sanchita Das[ab#], Varun Shukla[a], Lakshay Tyagi[a], Nishant Dafale[b,d]*, Sukdeb Pal[c,d*], Rita Dhodapkar[a,d]*

[a]Director's Research Cell, CSIR-National Environmental Engineering Research Institute (CSIR-NEERI), Nagpur – 440020, India

[b] Environmental Biotechnology and Genomics Division, CSIR-National Environmental Engineering Research Institute (CSIR-NEERI), Nagpur – 440020, India

[c]Wastewater Technology Division, CSIR-National Environmental Engineering Research Institute (CSIR-NEERI), Nagpur – 440020, India

[d]Academy of Science and Innovative Research (AcSIR), Ghaziabad- 201002, India

#Equal Contribution

* Corresponding authors:

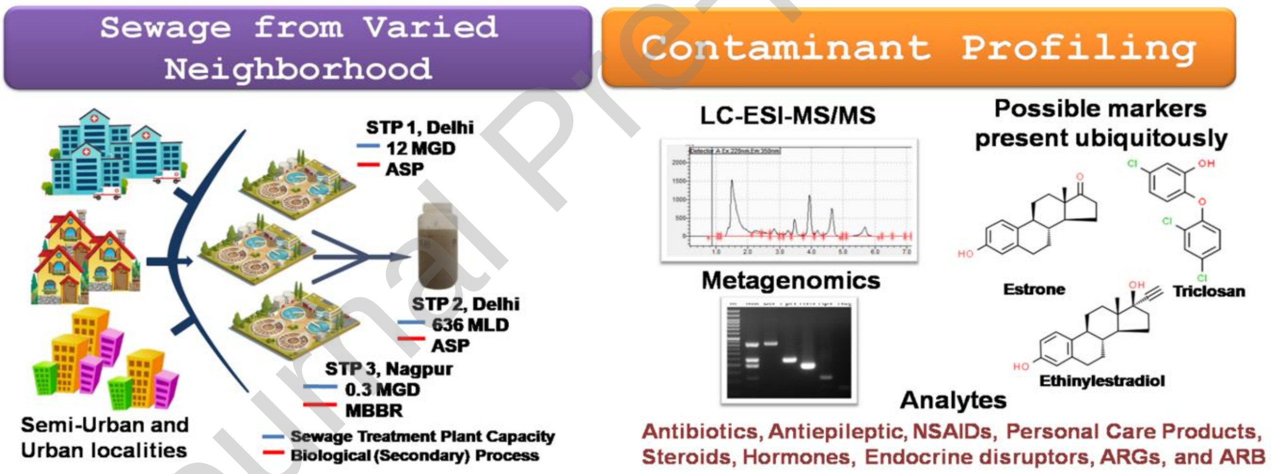
Tel.+91-712-2249885;

Emails: s_pal@neeri.res.in (SP); na_dafale@neeri.res.in (ND); rs_dhodapkar@neeri.res.in (RD);

Abstract:

In India, sewage (partially-treated/ untreated) is randomly used for irrigation because of easy availability and presence of residual organics and nutrients. However, data on the occurrence of contaminants of emerging concerns (CECs) such as pharmaceuticals and personal care products (PPCPs) and antibiotic resistant bacteria (ARB) and antibiotic resistant genes (ARGs) in sewage is scarce in Indian perspective. Herein, for the first time, we present a quantitative contamination profiling of selected PPCPs and antibiotic resistance in untreated and biologically-treated sewage from three different sewage treatment plants, located in northern and central part of India. Profiling of PPCPs were done using LC-ESI-MS/MS whereas antibiotic resistance was analyzed using gradient PCR and qPCR techniques. PPCPs were detected both in untreated and treated samples (0.4 – 1340 µg/L). A reduction in ARB and ARG load (2-3 log) and an increase in ARG copy number with respect to beta lactams and tetracycline were observed in treated sewage. Triclosan, estrone and 17α–ethynylestradiol, ubiquitous in all samples, could be used as markers for performance monitoring of sewage treatment facilities. The results obtained in this study help evaluate health and ecological risks associated with the presence of CECs in treated sewage used for irrigation and frame future policies.

Graphical abstract



Keywords: Sewage treatment plant, Pharmaceuticals and personal care products, antibiotic resistance, antibiotic resistant bacteria, gene quantification

1. Introduction

Contaminant of Emerging Concerns (CECs) are the set of natural and artificial chemical substances and their by-products or pathogens including antibiotic resistant bacteria (ARB) and antibiotic resistant genes (ARG) that occur in the water sources around the globe and are not currently monitored, but have ability to cause known or suspected detrimental effects on human health and ecosystem (Geissen et al., 2015, Menger et al., 2020). In recent years pharmaceuticals and personal care products (PPCPs) have gained attention as contaminants of emerging concerns (CECs) due to their potential threat to the environment and are considered to be causing environmental diseases of unknown etiology (Halden, 2015). The most widely identified PPCPs are the non-steroidal anti-inflammatory drugs, antidepressants, antibiotics, lipid metabolism drugs, hormonal agents, antiepileptic drugs, and β-blockers (Kowalski, 2011) and all types of cosmetics, including sunscreens (UV blockers), shampoo, soaps and antiseptics (Gałwa-Widera, 2019). PPCPs can enter the environment by excretion, domestic waste, sewage, bathing, and direct discharge in a variety of ways. Sewage treatment plant (STP) effluents, STP biosolids, agricultural and livestock activities are the primary sources of these emerging pollutants into water environment (Starling et al., 2019). Moreover, they have been widely found in stormwater runoffs, groundwater, and also in drinking water (Lo'pez-Serna et al., 2013). The existing conventional sewage treatment systems are shown not capable of removing PPCPs readily (Bhadra et al., 2017; Kasprzyk-Hordern et al., 2009). Incomplete or partial elimination of CECs poses a possible threat to the ecosystem. Recent studies have reported more than 100 CECs (Ebele et al., 2017, G Archana et al.,2017) identified with a concentration ranging from μg/kg to mg/kg in soil and ng/L to μg/L in groundwater, and surface water (Gałwa-Widera, 2019). As CECs are present in relatively low concentrations, direct qualitative and quantitative analysis of these chemicals in complex matrices such as wastewater is a challenging task.

Drug resistant pathogens have been identified as one of the key factors for the resurgence of contagious diseases. While microbial evolution is a natural phenomenon, improper and indiscriminate usage of antibiotics creates extra burden and promotes development of resistance (Allen et al., 2010; Andersson and Hughes, 2014). Due to the increasing resistance rate Antimicrobial Resistance (AMR) has emerged as one of the most prominent public health risks jeopardizing medicine in this century (O'Neill, 2014). Strong seasonal variation of ARGs within wastewater has been reported (Caucci et al., 2016). Higher levels of ARGs in Autumn and Winter were found to coincide with the rise in antibiotic prescribed in those seasons. Nonetheless, microbiological contaminants such as ARB and ARGs are still lesser known and not monitored in domestic wastewater (Bombaywala et al., 2020). Various metagenomic profiles of past investigations indicated STPs as hotspots for ARGs and their transfer amongst bacteria makes STPs direct source of spread of ARGs into the environment (Gupta et al., 2018; Dafale and Purohit 2016). The application of sewage for agricultural purposes is a common practice, and also a significant threat for antibiotic resistance dissemination to the environment (Sabri et al., 2020). Specific antibiotic resistance abundance patterns of various STPs targeted using advanced analytical techniques such as PCR and qPCR reveal ARGs and ARB resistomes that have evolved in the environment causing health risks in humans (Ren at al.,2018).

In India an estimated 62,000 million liters of sewage is generated every day, while the treatment capacity of the STPs installed across the country is only ~ 37% of sewage generated. Moreover, due to operation and maintenance issues many of the listed STPs are dysfunctional and many of them are operated at low service level compared to their design capacity. Thus, no more than 30% of sewage generated flows through treatment plants. Also, most of the STPs are not equipped with advanced tertiary treatment facilities and biological treatment (secondary treatment) is the penultimate treatment before disinfection. Thus, huge volume of untreated and inadequately treated sewage is directly or indirectly discharged in to the receiving environment every day (CPCB 2015). Because of easy and free availability, and presence of organics and nutrients untreated or partially treated sewage is a popular choice among farmers and is randomly used for irrigation purposes especially in water stressed areas. Such indiscriminate application of mostly untreated and partially treated sewage in agricultural activities potentially pose severe health risk. While presence of PPCPs in sewage could foster the spreading of ARGs which in turn may disseminate the resistance determinants potentially impacting the human health (Szekeres et al., 2018), irrigation using sewage contaminated with CECs may also result in to bio-accumulation of these compounds in agricultural produce leading to another route of human exposure (Miller et al., 2016). Unfortunately, to the best of our knowledge, till date there is no scientific and systematic data available on the presence of CECs in treated/untreated sewage in the Indian perspective. Therefore, the primary objective of this study was to identify and wherever possible quantify a set of selected PPCPs, along with ARGs and ARB in Indian STPs. Thus, in this study, we for the first time, examined the quantitative profiling of selected CECs including ARB and ARGs in sewage sampled from different STPs in India. Detection of these contaminants in the environment depends heavily on its geo-spatial and temporal variables, regional economy, consumption of drugs, STP facilities, and local conditions. To encompass this variability, we selected STPs that cater to sewage generated from varied neighborhood of different geographical and climatic conditions- urban area in northern India (STP1 and 2) and semi-urban area in central part of India (STP 3). The contamination profiles of selected PPCPs, ARGs and ARB were assessed in influent and secondary-treated effluents of three different STPs. Selected PPCPs consisting of antibiotics (ciprofloxacin & sulfamethoxazole), antibacterial (triclosan), antiepileptic (carbamazepine), non-steroidal anti-inflammatory drug (ibuprofen), steroids and hormones (estrone, 17-α ethynylesdtradiol, testosterone and progesterone), endocrine disruptors (bisphenol-A) and psychoactive drug (caffeine) were screened for their occurrence using Ultra-Fast Liquid Chromatography (UFLC) and the findings were further corroborated though LC-MS analysis. Abundance profile of ARB and ARGs were analyzed in STP influent, effluents as well as in accumulated sludge and in water used for irrigation using gradient PCR and qPCR techniques.

<Figure 1>

2. Materials and Method

2.1 Location of STPs and sewage sampling

Three different STPs from different geographical areas were selected for mapping the emerging contaminants which depends heavily on its geo-spatial and temporal variables, regional economy, consumption of drugs, STP facilities, and local conditions (**Figure 1**). STP 1 and STP 2 represent urban catchments in Delhi (northern India). STP 1 is designed to treat 45 MLD (million liters per day) of sewage while the treatment capacity of STP 2 is 636 MLD of sewage. Biological treatment in both plants is operated based on activated sludge process (ASP). Secondary-treated effluent from STP 1 is discharged in to an irrigation canal flowing by the STP. STP 3 in Nagpur (central India) is comparatively a smaller treatment plant that treats around 0.3 MLD of sewage from a semi urban catchment and operates on moving bed bioreactor (MBBR) technology. Grab samples (1 L) were collected from five different locations of equalization basin as well as from the treated effluent collection sump, placed in a clean acid-rinsed polypropylene bucket, mixed well and subsequently 1 L of this well-mixed sample was taken to make 1 L of space composited sample. Additionally, surface water samples were also collected from the irrigation canal at ~100 meter downstream from the point of discharge of treated effluent from STP 1. The canal also receives treated sewage from 2 other adjacent STPs. Water from this canal is directly used for irrigation of cash crops like vegetables and paddy in the adjacent agricultural fields. All samples were placed directly into acid-rinsed polypropylene containers without any filtration. Parameters such as pH, temperature, and dissolved solids were monitored onsite. All other analyses were carried out in laboratories at CSIR-NEERI, Nagpur, India. Samples were preserved immediately after collection. For analysis of CECs collected samples were preserved at pH 2. For microbiological analysis the samples were collected in sterilized containers. Samples for CEC and microbiological analysis were stored at 4°C until further analysis. The samples collected for physicochemical characterization were preserved, processed and analyzed for different parameters (total suspended solid (TSS), chemical oxygen demand (COD), biochemical oxygen demand (BOD), total phosphorus (TP), total Kjeldahl nitrogen (TKN) etc.) according to standard methods (APHA 2017).

2.4 Sample preparation and solid phase extraction (SPE)

All samples and reagent blanks (1000 mL each) were first centrifuged at 4500 rpm for 10 mins and subsequently filtered through Whatman Glass Microfiber filters (70 mm GF/C filter; pore size 1.2 μm) prior to the solid phase extraction (SPE). The filtrates were extracted using SPE column (Waters® Oasis C18 cartridge). Cartridges were preconditioned with 10 mL of distilled water, 10 mL of methanol, and subsequently rinsed with distilled water. Samples were passed through the pre-conditioned cartridge at a flow rate of 2 mL/min. Cartridges were dried under vacuum for 20 min after extraction. Subsequently, the analytes of interest were eluted sequentially using 8 mL methanol and then 8 mL of methanol containing 0.1% acetic acid. The mixture of methanol/acetic acid was evaporated to dryness under nitrogen gas before reconstitution with methanol (1 mL) and filtered through a PTFE syringe filter (pore size 0.02 µm) before subjecting to UFLC and LC/ESI-MS/MS analysis. Recovery of the SPE procedure was determined by spiking the samples with appropriate amounts of analytes prior to extraction and in 1 mL of the reconstituted extracts.

2.5 Target PPCPs and Instrumental analysis

In total, eight pharmaceuticals (ciprofloxacin, sulfamethoxazole, carbamazepine, ibuprofen, 17-α ethynylesdtradiol, estrone, testosterone and progesterone), three personal care products (triclosan and bisphenol-A) and one central nervous system stimulant (caffeine) were analyzed in the collected sewage filtrates. Selection of these analytes were primarily influenced by their availability in the Indian market. The qualitative as well as the quantitative analysis was confirmed by both UFLC and LC-(ESI) MS-MS.

2.5.1 UFLC conditions

Initial separation and screening of the PPCPs were carried out using an ultra-fast liquid chromatograph (Prominence UFLC, Shimadzu, Japan) coupled with a fluorescence detector (RF-20A), and a photodiode array (PDA) detector. A reverse-phase RP-18(5 µm) column was used for the separation of analytes in the processed sewage sample. The column was kept at 27°C. 10 µL of each sample was injected with the run time of 40 min. The chromatographic analysis was performed using two sets of binary mobile phases; Method 1: formic acid (0.1%) in HPLC grade water (Solvent A)-acetonitrile (Solvent B) (40/60; v/v) and Method 2; water (Solvent A)-methanol (Solvent B) (20/80; v/v) both containing 0.1 M ammonium acetate and 0.01% formic acid. Prior to each separation the column was equilibrated with the respective mobile phases for 3 min. Samples were eluted using simultaneous mobile phase and flowrate gradient. Elution started with 100% of A maintained for 6 min, followed by a linear gradient to 50% of eluent B for 19-min and 5-min linear gradient back to 100% of A. Flow rate gradient started with 1.0 mL/min for 6 min, followed by a linear gradient of 0.8 mL/min for 19- min and a linear gradient back to 1 mL/min for 5-min. The target PPCPs were eluted within 20 min. The column was re-equilibrated for 20-min.

2.5.2 LC-(ESI) MS-MS conditions

MS analyses were carried out using a triple quadrupole liquid chromatograph mass spectrometer (LC/ESI-MS/MS 8040, Shimadzu, Japan). The MS analyses were carried out simultaneously in the positive and negative Electron spray ionization [ESI (+) & ESI (−)] modes. Nitrogen was used for collision-induced dissociation at approximately 230 kPa. Targeted analytes were monitored in multiple reaction monitoring (MRM) and scan mode. Analytes were quantified based on an external calibration established using freshly prepared standards (1 – 8 µg/mL). The electro spray source block temperature was adjusted at 300 °C while desolvation temperature was kept at 250 °C. Drying gas flow rate was 15 L/min and nebulizing gas flow rate was 3L/min. Isotopically labelled internal standards were used for evaluation of matrix effects. The extracts were spiked with these standards immediately prior to the LC-MS analyses for matrix control. Samples were diluted and reanalyzed to minimize the matrix effect when the signal dropped below the threshold (taken as 60% response in calibration).

2.6 Method validation and quantification

A pilot study was conducted to establish the chromatographic conditions, linearity, selectivity, limit of detection (LOD) and limit of quantification (LOQ) etc. To examine any possible instrumental and sample contamination that could possibly influence the detection and quantification, blank control samples (n=3; SPE cartridges without any field samples) were prepared under identical laboratory conditions as the field samples. For example, several HPLC grade water blanks were processed though SPE cartridges and analyzed to determine any possible cross contamination during sample preparation. Concentrations of analytes in these blanks were below the respective LOD. Additionally, solvent blanks were also analyzed at regular interval (every ten injections). Carryover of the target compounds/ mass-labeled targets was not detected. Intermediate QA/QC was checked after every 10 samples to examine LC-MS instrument sensitivity. For each analyte MRM transitions between the precursor ion and two most abundant fragment ions were monitored. Quantification was done using the highest characteristic MRM transition between precursor /product ion. The other one was used for confirming the presence of target analytes in the samples except for the analytes with poor fragmentation. Additionally, the LC retention time (retention factor) of analytes was also compared with the respective reference/internal standards for detection of the analytes in the field samples. Calibration curves were constituted analyzing the standards. A linear least-squares regression analysis was used to calculate the linearity of the method in the studied concentration range. Samples spiked with analytes at appropriate concentrations (before SPE and in the reconstituted 1 mL extract) were used to establish the recovery of the methods.

The limit of detection (LOD) and limit of quantification (LOQ) of the methods were defined and calculated as the lowest discernible concentration of the individual compound from spiked water extracts in MRM mode at a signal-to-noise ratio (S/N) of 3 and 10, respectively. LOD and LOQ for the method were determined using standards prepared by serial dilution until the signal-to-noise ratio reaches the value of 3 and 10 for LOD and LOQ, respectively (27). Molecular weight, LC retention time, LOD and LOQ as determined using LC-ESI-MS/MS of individual analytes are summarized in **Table 1.**

< Table 1>

2.7 Statistical analysis

Statistical analysis system (SAS) software (Version 9.1) was used for one-way analysis of variance (ANOVA) and Duncan's multiple range tests to determine the significant difference between the mean values of control and experiment groups. The results were considered to be significant at $p < 0.05$. OriginPro 8.5 software was used for graphing.

2.8 Microbiological and molecular analysis

For microbiological and molecular analysis, commercially available M-EC, M-FC, M-Endo, Luria Agar, antibiotics (ciprofloxacin, ampicillin, tetracycline, sulfamethoxazole, chloramphenicol), TaKaRa PCR kits containing buffer, dNTP, taq polymerase enzyme were used. For qPCR quantification forward and reverse primers (Eurofins) and SYBR Green dye (Applied Biosystems) were procured.

2.8.1 Enumeration of indicator bacteria and ARB on selective mediums

In this method, unfiltered water samples were serially diluted, preferably to obtain separate isolated bacterial colonies while plating, in the range of 30 to 300 cells per plate. An aliquot of 1 mL of diluted sample was spread evenly on a M-EC, M-FC, and M-Endo agar plates for the detection of *E. coli*, fecal coliform, and total coliform, respectively. All plates were incubated at 37^{o}C for 48 hours. Number of bacteria in the sample (CFU/100 mL) was calculated from the number of colonies counted on the plates. Antibiotic resistant bacteria were enumerated on the antibiotic containing Luria Agar plates as reported previously (Turolla et al., 2018). Antibiotic concentrations were selected according to the EUCAST breakpoint table (EUCAST). ARBs were then counted on Luria Agar plates supplemented with selected antibiotics.

2.8 Antimicrobial resistance detection through gene specific gradient PCR method

For molecular analysis, metagenomic DNA extraction was conducted using Fast DNA Spin kit for soil (MP Biomedicals, CA, USA). DNA extraction was carried out following the manufacturer's guidelines, and its purity was checked using microspectrophotometry (DeNovix DS-11 Spectrophotometer). The extracted metagenomic DNA was then used for qualitative PCR analysis to check the presence of targeted ARGs blaTEM, tetM, sul2, int1, qnrS, blaCTXM, oqx, parC and only the ARGs that were detected in samples through gradient PCR, were further for qPCR analysis. For gradient PCR analysis, amplification of template DNA was done with commercially available kits (TaKaRa) according to manufacturer's protocol. Each 25 µL of reaction mixture contained a buffer for PCR, dNTPs, forward and reverse primers, DNA template. Each PCR cycle commenced with an initial 5 min DNA denaturation step at 95^{o} C, followed by 35 amplification cycles each consisting of 1) denaturing for 30 sec at 95^{o} C; 2) annealing for 30 sec to 1 min at the selected annealing temperature for all ARGs; 3) extension for 2 min at 72^{o} C. A final 7 min extension at 72^{o} C was performed.

2.9 Quantification of ARGs present in STPs

Detection of ARGs was performed using suitable PCR primers, designed and synthesized through Eurofins, India. Genomic DNA of known concentrations carrying target genes were used to generate standard calibration curves. The concentration of DNA carrying target genes was determined using nanodrop. Samples prepared by serial dilution of known copy numbers of DNA were used to plot the standard curve which was used to determine the ARG copy numbers in STP samples (Zhang et al., 2020). ARGs were detected and quantified using a Bio-Rad CFX96 instrument. A quantitative PCR reaction was executed in a 25 µL volume mixture and was run in 96-well microplates containing 10 µL of SYBR green premix (Applied Biosystems). The following protocol was followed: 30s at 95^{o} C, followed by 40 cycles of 5s at 95^{o} C, 30s at respective annealing temperature, extension for 30s at 72^{o} C, then a final melt curve stage with temperature up to 90^{o} C$^{.}$ Real time PCR is an advanced tool for the quantification of genes. The threshold cycle (CT) is the endpoint of quantification for real-time qPCR. The CT value is defined as the amplification cycle at which the reporter dye crosses the threshold point and releases a fluorescent signal that is employed. The data presented in CT value ensures the phase of exponential amplification in qPCR. The CT value is inversely proportional to the number of target genes present in the sample (i.e., the lower the CT value, the higher the amount of the target gene). For ARG fold change interpretation in influent and effluent of wastewater samples, Livak's method for relative qPCR quantification method was used (Schmittgen and Liyak et al., 2008; Stange et al., 2019), the following calculations were used: Fold change = 2 $^{-\Delta\Delta CT}$;

where,

$$2^{-\Delta\Delta CT} = [CT_{gene\ of\ interest} - CT_{internal\ control}]sample\ A - [CT_{gene\ of\ interest} - CT_{internal\ control}]sample\ B$$

3. Results and Discussion

3.1 Physicochemical characteristics of sewage samples

Physicochemical characteristics of the treated sewage samples reveal that total suspended solids (TSS) ranged between 1-23 mg/L. COD and BOD of the effluents varied between 30-126 mg/L and 8-50 mg/L, while total Kjeldahl nitrogen (TKN) and phosphate-phosphorous ranged between 1.0-37.5 mg/L and 2.4-8.4 mg/L, respectively. STP 3 did not meet the standards for discharge after secondary treatment of MBBR. The comparative study of STPs referring to the treated sewage norms laid by the Hon'ble National Green Tribunal (NGT), India (NGT Order 2019) (Table **S1**) shows that the biochemical oxygen demands (BOD) of treated effluents from STP 1 & 2 are within the permissible limit. The irrigation canal that carries mixed effluents from STP1 and other two STPs had a TSS of 76 mg/L and COD of 90 mg/L that are higher than the stipulated standards for disposal of treated sewage on to land. Standards for discharge with respect to nutrients (TKN and TP) could not be met by any of the STPs.

3.2 Optimization of HPLC analysis

In chromatographic separation, role of mobile phase is very important for effective separation of compounds (Kasprzyk et.al., 2007). To evaluate the role of mobile phases, two sets of binary solvents were used as mentioned earlier (2.5.1 UFLC conditions). The selection of method was based on the detection of the targeted compounds, their separation and signal intensity. While bisphenol-A, triclosan, Ibuprofen, caffeine, sulfamethoxazole, ciprofloxacin and carbamazepine were detected better using method 1, analytes like 17-α-ethynylesdtradiol, estrone, testosterone, progesterone were detected using method 2.

<Figure 2>

3.3 Profiling of PPCPs

The collected influent and effluent samples from three different STPs were analyzed for PPCPs using UPLC-LC-(ESI)-MS/MS. Two different analytical methods were used to mitigate the biasness involved in them. The quantitative analysis of CECs carried out using LC-MS/MS reveal that, the targeted PPCPs were present in varying concentrations at all the STPs. Presence of overall ten CECs was detected in STP1 (influent & effluent). The method 1 and 2 could detect 6 and 4 CECs, respectively (**Figure 2)**. Similarly, in STP 2 presence of 7 and 2 CECs was detected using method 1 and 2, respectively (**Figure 3)**. Interestingly, in case of STP 3, method 1 and method 2 could detect 3 CECs each (**Figure 4**). The rest of the target CECs were found below the detectable limit in this STP located in the semi-urban area of central India. This could, though arguably, be attributed to the different usage pattern of the pharmaceutical and personal care products in geo-spatially different urban and semi-urban areas as well as to the difference in population that these STPs cater.

<Figure 3>

Out of eleven targeted analytes, almost all of them were detected in the influent samples of one or more STPs. The Toxic Unit (TU) of an analyte was obtained by normalizing the concentration value for an analyte with its reported LD_{50} value. The sum of the Toxic Units of the detected PPCPs in the influent samples (ΣTU-PPCPs$_{in}$) of all the monitored STPs ranged between 0.017 and 0.004. The highest ΣTU-PPCPs$_{in}$ was found in STP 1 while the lowest was detected in STP 3. Sulfamethoxazole was found in the highest concertation whereas triclosan was detected to have the lowest concentration among the CECs. Triclosan, ubiquitous in all the sample, was detected to have low concentration due to fast photodegradation which led to the formation of number of metabolites (Prado et al., 2006).

Similarly, effluent samples confirm the presence of nine PPCPs at all STPs. The sum of the Toxic Units of PPCPs detected in the effluent samples (ΣTU-PPCPs$_{eff}$) ranged between 0.002 and 0.017. The highest ΣTU-PPCPs$_{eff}$ was found in STP 3 while the it was lowest in STP 2. Ciprofloxacin was detected in the highest concertation whereas triclosan was detected to have the lowest concentration among the CECs in treated effluents.

<Figure 4>

The variation in the concentrations of the PPCPs in the samples as found in this study could be attributed to the varying concentration of these PPCPs in different products and variation in the consumption pattern of such products. The concertation of PPCPs in effluent samples were obviously lower than that found in the corresponding influent, which suggests that the existing conventional wastewater treatment facility is partially capable of removing these contaminants. However, in some cases, concentrations of compounds like carbamazepine, ibuprofen, and ciprofloxacin in effluent were higher than influent. Higher concentration of these PPCPs in effluent samples could be either because of their poor degradation or the due to deconjugation of inactive conjugates of these compounds during the treatment process increasing the release and hence concentration of the active parent compounds in effluent (Kirk et al. 2002; Andersen et al. 2003; D'Ascenzo et al. 2003). Other possible reason of the higher effluent concentrations of these PPCPs may simply be the diurnal variations of these compounds. The sampling may not necessarily represent the peak load. Although both influent and effluent samples were collected during the same period, downstream samples of the influent represent conditions of previous time which is close to the hydraulic retention time of the STP. (Karnjanapiboonwong et al., 2011). It is noteworthy that triclosan, 17-α ethynylestradiol (EEA), and estrone were detected both in influent and effluent samples of all

STPs, suggesting that they could be used as signature compounds to monitor the presence of PPCPs in sewage as well to assess the performance efficiencies of the wastewater treatment facilities in the STPs.

The removal of the target contaminants in the secondary treated effluent with respect to their concentrations in influent samples showed a high variability at different STPs. While some compounds like bisphenol-A, 17 α–ethynylesdtradiol, triclosan and progesterone were significantly reduced (more than 63%) in the treated the effluent, compounds like ibuprofen, estrone, caffeine, sulfamethoxazole, testosterone and ciprofloxacin were reduced by less than 40%. Lower / negative treatment efficiencies of carbamazepine has been reported earlier by aerobic biological processes (Guohua et al., 2014; Nakada et al., 2010). Similarly, in our studies removal efficiencies in the range of 1.8% to -5% was observed for carbamazepine. Carbamezapine has a low K_{ow} of 2.4 and is expected to be present in effluent because it is not easily absorbed into organic matter or sewage sludge, resulting in resistance towards microbial degradation (Archana et al., 2017). It is found that excessive concentration of carbamazepine can cause toxic epidermal necrolysis, although it behaves as anticonvulsants for treatment purposes (Hai et al., 2018). In the case of ciprofloxacin, the removal efficiency was found to be nearly insignificant in STP 2 as well as in STP 3. There are many different combinations of factors such as the structure of the compound, low Kow, and properties that could affect the removal of these compounds by resisting their susceptibility to settling and biodegradation (Rahul et al., 2019).

<Figure 5>

3.4 Detection of indicator bacterial load and *E. coli* in effluents

Persisting contamination of indicator bacteria was detected in STP 1 effluent, sludge, and irrigation canal samples. Though the load was removed by 3.4 log in the effluent, 3 log in sludge, and 2.3 log in irrigation canal samples, bacteria still endured the treatments. Total coliform as high as 22.4×10^6 was reduced to 8×10^4 CFU/100 ml, out of which faecal coliform and *E. coli* were 3.6×10^2 and 2.4×10^2 CFU/100 ml respectively. STP 1 effluent leads to an irrigation canal, which supplies the treated wastewater for irrigation purposes. Whereas, is STP, total coliform was removed by 5 log. Hence 8.3×10^3 CFU/100 ml was detected in the effluent. In STP 3 colony STP effluent, 2.1×10^3 CFU/100 ml total coliform was found, out of which 5.1×10^2 was *E. coli*. Hence, only 2 log of total coliform were removed from their influent (**Figure 5**).

The indicator bacterial load found in the effluent of selected STPs mark the presence of potential contamination persisting in treated wastewater. Delanka-Pedige et al., (2019) showed removal of 3.3 log of 10^7 CFU/100 ml in treated wastewater, which, according to WHO guidelines (2017), was unsuitable for reuse. In this present study, STP 1 treated effluent showed the presence of more than 10^5CFU/100 ml of total coliforms, which according to WHO guidelines, are not suitable for reuse and require post-tertiary treatment for the removal of microbial load. Treated sewage effluents containing microbial contaminants are often discharged, which can contribute to the prevalence of pathogens in the environment (Maharjan et al., 2018). Coliforms are a group of bacteria that are potential indicators of microbial contamination and water quality (Zhao et al., 2019). *E. coli* is one of the important bacterial indicators of fecal contamination in wastewater. This study showed that the bacterial load is frequently discharged into the environment through STPs. Since wastewater from hospitals, households, and pharmaceutical industries usually get collected in sewage that flows into STPs, high fecal bacterial load was expected in the influent of sewage (Naquin et al., 2017). Presence of total coliforms and *E. coli* in treated sewage from all STPs showed that the treatment processes and disinfection techniques used at plants are ineffective in removing fecal bacteria, implying that the treated effluents from STPs is of poor microbiological quality (Li et al., 2017).

<Figure 6>

3.5 The detection of antibiotic resistance bacterial load in STPs

The influent and effluent samples of STPs were screened for the presence of antibiotic resistant bacteria for commonly used antibiotics (ampicillin, tetracycline, sulfamethoxazole, ciprofloxacin and chloramphenicol). STP 1 influent samples gave a high number of ampicillin resistant bacteria, i.e., 15.7×10^5 CFU/100 ml, which was reduced to 2.4×10^2 CFU/100 ml in the treated effluent. Sulfamethoxazole and ciprofloxacin resistant bacteria were found to be 2×10^2 and 5×10^2 CFU/100 ml in effluent samples. Tetracycline and chloramphenicol resistant bacteria were not detected in effluent samples; however, they were present in sludge samples comprising of 3.6×10^2 and 4.1×10^2 CFU/100 ml (*Figure 6*). STP 2 influent consisted 12×10^5 CFU/100 ml of ampicillin resistant bacteria, reduced to 13.5×10^2 CFU/100 ml in the effluent. Subsequently, sulfamethoxazole and tetracycline resistant bacteria comprised 6.1×10^2, 4.3×10^2 CFU/100 ml of the effluent sample. However, a high number of ciprofloxacin and chloramphenicol resistant bacteria were found, comprising of 8.6×10^2 and 8.8×10^2 CFU/100 ml of treated effluent. STP 3 effluent had a comparatively low ARB count. Influent comprising of 6.8×10^4, 5×10^3 and 6.1×10^3 CFU/100 ml of ampicillin, sulfamethoxazole, and chloramphenicol resistant bacteria respectively, which was reduced to 5.2×10^2, 12×10^2 and 2.2×10^3 CFU/100 ml of effluent. Also, a high number of ciprofloxacin and tetracycline resistant bacteria were found in influent for 9.5×10^3 and 8.3×10^3 CFU/100 ml, which was reduced by 2 folds in the effluent (**Figure 6**).

In this study, the occurrence of ARBs pattern in three different STPs in India whose effluents have been utilized for irrigation purposes was investigated. The results show that the treated effluents cater resistant bacteria to some commonly used antibiotics (ampicillin, sulfamethoxazole, ciprofloxacin, tetracycline, and chloramphenicol) (Naquin et al., 2015). The resistant bacteria to these antibiotics were also observed in the irrigation canal sample, which directly supplies treated water for irrigation purposes. Effluent water showed a significant reduction in ARBs by 10^2 CFU/100 ml, but as the bacterial species have a peculiar tendency to horizontally transfer genes (HGT), it becomes difficult to control the spread of ARGs in the environment. The results show the prevalence of ampicillin resistant bacteria in influent samples, as it is one of the most extensively used antibiotics, which suggests the resistance that is developed in the course of time (Timraz et al., 2017). In the present study, we have mapped ARGs, ARBs, as well as antibiotic residues present in the samples. Some of the common compounds targeted such as, sulphamethoxazole and ciprofloxacin, confirm their presence through chromatography as well as their respective resistant bacterial counts. In the case of sulphamethoxazole, a very low removal (6 μg/L) was found in STP 1 and 2 effluent samples, while its resistant bacteria were detected in all STPs with a low count. However, for ciprofloxacin, which marks its presence in STP 2 and 3 with a high concentration in both influent and effluent samples, its resistant bacteria were found to be present in all the STPs, but the bacterial count was minimal.

Multiple disinfection methods that are used in these STPs during the reclaimed water production significantly decreased the number of ARB (Aslan et al., 2018). Due to release of antibiotic residues in sewage and also the presence of other compounds such as, heavy metals and biocides such as quaternary ammonium salts, the selective pressure for bacterial communities for survival increases thus, a breeding hotspot emerges, where different resistance mechanisms are developed in bacteria, which leads to their newly formed resistant strains (Xu et al., 2014). However, there are currently no guidelines to monitor antibiotic resistant bacteria in reused water, which makes it tough to conclude whether the number of ARB found in the effluent sample makes the water safe for reuse.

<Figure 7>

3.6 Absolute quantification of ARGs in STPs

Target bands of resistance genes were detected by qualitative PCR-electrophoresis. The qualitative PCR results showed that target genes were positive. No non-specific bands were observed. Only the ARGs that were found in qualitative PCR were further quantified by qPCR analysis. ARGs selected for detection in this study include, tetracycline (tetM), beta lactamse (blaTEM), quinolone (qnrS) and sulfonamide (sul2). A few antibiotic specific resistant genes like blaCTXM (cephalosporin), oqx (ofloxacin), parC (ciprofloxacin) were also targeted to detect resistance genes against specific antibiotics. A reduction of 1 log was observed in sul2 genes in STP1 effluent, and the water used for irrigation had a 0.5 log reduction. In STP 2, 3.21×10^3 gene copies/ml were found in influent and 2.7×10^2 copies/ml in effluent samples, which shows around 1 log reduction in the effluent. In STP3, 2.1×10^6 copies/ml in influent was reduced to 10^2 copies/ml in the effluent, which shows a 4-log reduction of sul2 genes. However, it was observed that though the ARG copies were reduced in effluent samples, complete removal of sul2 genes was not achieved in any of the STPs. It was observed while sampling that, both treated and untreated wastewater get mixed in the canal that supplies water for irrigation from STP1. Hence, an increase in blaTEM and other ARGs, ARBs is observed in irrigation canal sample. Whereas, no reduction in the blaTEM gene copy number was seen in STP 3. Increased tetracycline resistance gene (tetM) copy number was detected in STP 2 by 10^2 copies/ml. There was no significant reduction in qnrS gene in all three STPs, as gene copy number was persistent in all samples. ARG copies/mL of blaCTXM, oqx gene was persistent in all samples, and very low reduction in effluent samples was detected. In STP 3, parC gene copy number was reduced by 10^2 copies/ml in effluent samples (**Figure 7**). The presence of the int1 (class 1 integron) gene in samples suggests mobile genetic elements play a significant role in horizontal gene transfer (HGT) events, which could develop a breeding hotspot for the emergence of ARB.

3.7 Relative quantification of ARGs present in treated effluents

An additional relative quantification method (Livak's method) was used to see the fold change pattern in ARGs present in wastewater samples. In this method, the ARG abundance was normalized with reference to 16s rRNA gene, and influent samples that were untreated wastewater were taken as a control for all three STPs (Figure 8).

A decrease in sul2 gene by 2.28 folds and an increase in blaTEM was observed in STP 3 colony effluent sample. Whereas, a decrease of 1.14 was detected in STP 3 colony sludge sample. In STP 2 samples, a decrease in the sul2 gene by 4.06 folds was remarkable, considering a drastic increase in tetM by 5.12 folds. A peak in the blaTEM gene by 6.02 folds in STP 1 effluent was also significant. Also, a decrease in int1 and sul2 genes by 1.09 and 2.56 folds, respectively, in STP 1 effluent is observed. Whereas irrigation canal effluent showed A decrease in blaTEM and sul2 by 4.08 and 2.14 folds, respectively. The qPCR abundance profile for these genes shows a reduction in effluents, but genes still persist.

4. Conclusion

The method used for the simultaneous determination of targeted CECs in different STPs confirmed the presence of pharmaceuticals including NSAIDs, hormones, and EDCs quantitatively in all secondary-treated sewage samples. Amongst the STPs, the highest numbers of contaminants were detected at STP 2, followed by STP 1 and STP 3. The observed CECs were found to have a concentration of µg/L at both influent and effluent. The study also concludes that amongst the targeted emerging contaminants, triclosan, estrone and 17 α – ethynylestradiol were found to be ubiquitous in all STPs. The persistent PPCPs in the effluent and in sludge marked the presence of ARB and ARGs and load of indicator coliforms in effluent suggests that the treated wastewater from all three STPs are not suitable for reuse. The qPCR results indicate a decrease in ARG copies in the effluent, but resistance genes still persist. This suggests that there is a need to employ a post-tertiary wastewater treatment to increase the efficiency of the STPs in the removal of ARB and ARGs. The physicochemical parameters except nutrients (total phosphate and TKN) of effluent samples were within the prescribed national treated effluent norms in STP 1 and 2. In the case of STP 3 most of the parameters exceeded the norms for discharge into surface water. Inadequacy of performance of STPs to remove nutrient parameters and residual harmful organics like CECs requires upgradation of the STPs and further polishing treatment before discharge. The data provided here could be useful for understanding the further improvement required within the wastewater treatment framework especially in terms of making treated sewage free of CECs (including ARB, ARGs) and fit for sustainable reuse.

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Figure legends

Figure 1 Location of STPs. The selected STPs cater to sewage generated from varied neighborhood of different geographical and climatic conditions- urban area in northern India (STP1 and 2) and semi-urban area in central part of India (STP 3).

Figure 2 (a) Detection of CECs in influent, effluent samples of STP 1 and nearby irrigation canal water using Method 1 [formic acid (0.1%) in HPLC grade water (Solvent A)-acetonitrile (Solvent B) (40/60; v/v)] (b) Detection of CECs in influent, effluent samples of STP 1 and nearby irrigation canal water using Method 2 [water (Solvent A)-methanol (Solvent B) (20/80; v/v) both containing 0.1 M ammonium acetate and 0.01% formic acid], *$p < 0.05$ vs control. Columns marked with the same uppercase letter are not significantly different for individual analytes among the group (influent/effluent/irrigation) by Duncan's multiple range test in one-way ANOVA ($p < 0.05$).The number on the column bar denotes the average concentration value (μg/L) of at least three-independent measurement of the samples (influent, effluent and irrigation). The error bar signifies ± standard deviation of at least three-independent measurements.

Figure 3 (a) Detection of CECs in influent and effluent samples of STP 2 using Method 1 [formic acid (0.1%) in HPLC grade water (Solvent A)-acetonitrile (Solvent B) (40/60; v/v)] (b) Detection of CECs in influent and effluent samples of STP 2 using Method 2 [water (Solvent A)-methanol (Solvent B) (20/80; v/v) both containing 0.1 M ammonium acetate and 0.01% formic acid], $*p < 0.05$ vs control. Columns marked with the same uppercase letter are not significantly different for individual analytes among the group (influent/effluent) by Duncan's multiple range test in one-way ANOVA ($p < 0.05$). The number on the column bar denotes the average concentration value (μg/L) of at least three-independent measurement of the samples (influent and effluent). The error bar signifies ± standard deviation of at least three-independent measurements.

Figure 4 (a) Detection of CECs in influent and effluent samples of STP 3 using Method 1 [formic acid (0.1%) in HPLC grade water (Solvent A)-acetonitrile (Solvent B) (40/60; v/v)] (b) Detection of CECs in influent and effluent samples of STP 3 using Method 2 [water (Solvent A)-methanol (Solvent B) (20/80; v/v) both containing 0.1 M ammonium acetate and 0.01% formic acid], $*p < 0.05$ vs control. Columns marked with the same uppercase letter are not significantly different for individual analytes among the group (influent/effluent) by Duncan's multiple range test in one-way ANOVA ($p < 0.05$). The number on the column bar denotes the average concentration value (μg/L) of at least three-independent measurement of the samples (influent and effluent). The error bar signifies ± standard deviation of at least three-independent measurements.

Figure 5 Number of indicator bacteria in influent, effluent and sludge samples collected from different STPs. Indicator bacteria count in irrigation canal water near STP1 is also shown.

Figure 6 Antibiotic resistant bacteria count in influent, effluent and sludge samples collected from different STPs. Antibiotic resistant bacteria count in irrigation canal water near STP1 is also shown.

Figure 7 Absolute quantification of ARGs influent, effluent and sludge samples collected from different STPs and that in irrigation canal water near STP1.

Figure 8 Normalized values of the tested ARGs in effluent and sludge samples collected from different STPs. The ARG abundance was normalized with reference to 16s rRNA gene, and untreated wastewater (influent samples) was taken as a control for all three STPs.

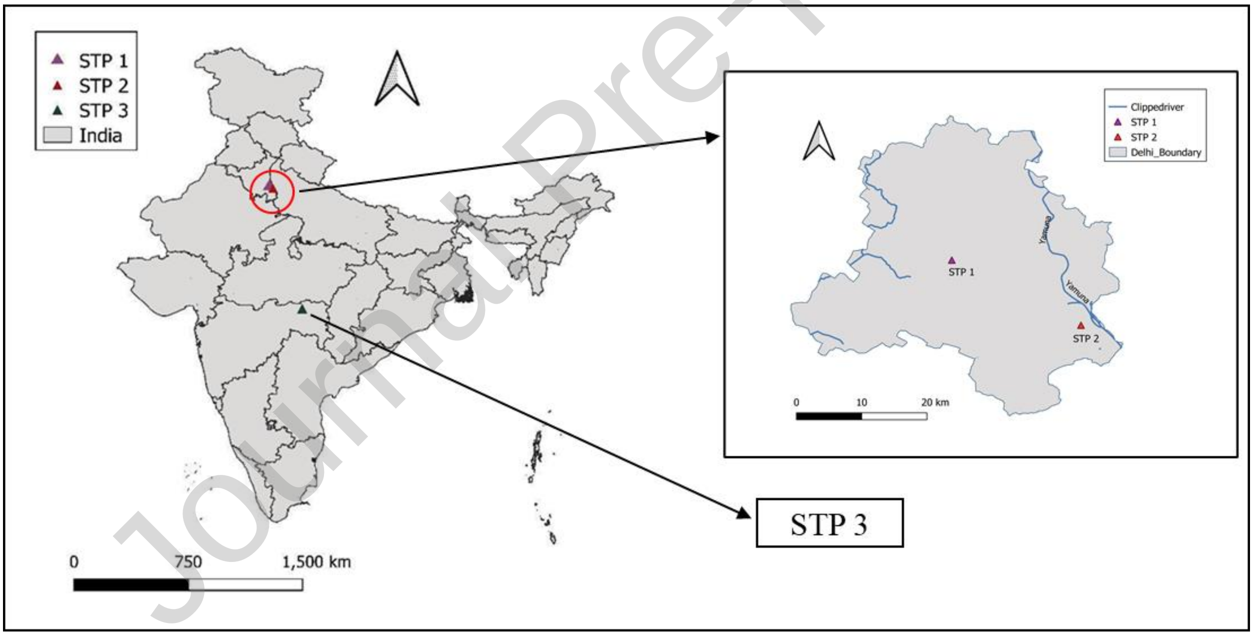

Figure 1

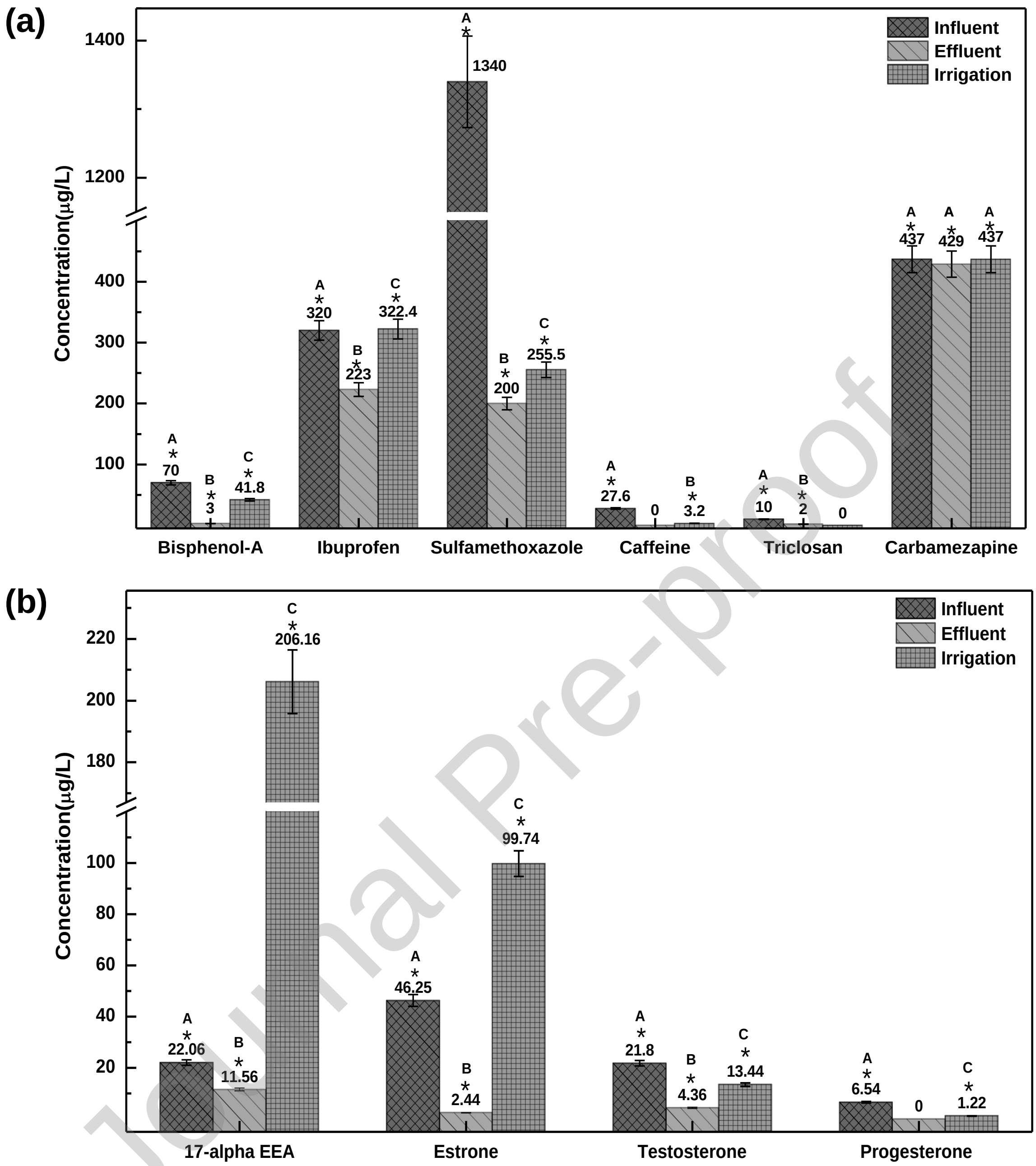
(a)
Influent
Effluent
Irrigation
Concentration(μg/L)
1400
1200
400
300
200
100
A
70
B
3
C
41.8
A
320
B
223
C
322.4
A
1340
B
200
C
255.5
A
27.6
0
B
3.2
A
10
B
2
0
A
437
A
429
A
437
Bisphenol-A
Ibuprofen
Sulfamethoxazole
Caffeine
Triclosan
Carbamezapine
(b)
Influent
Effluent
Irrigation
Concentration(μg/L)
220
200
180
100
80
60
40
20
A
22.06
B
11.56
C
206.16
A
46.25
B
2.44
C
99.74
A
21.8
B
4.36
C
13.44
A
6.54
0
C
1.22
17-alpha EEA
Estrone
Testosterone
Progesterone

Figure 2

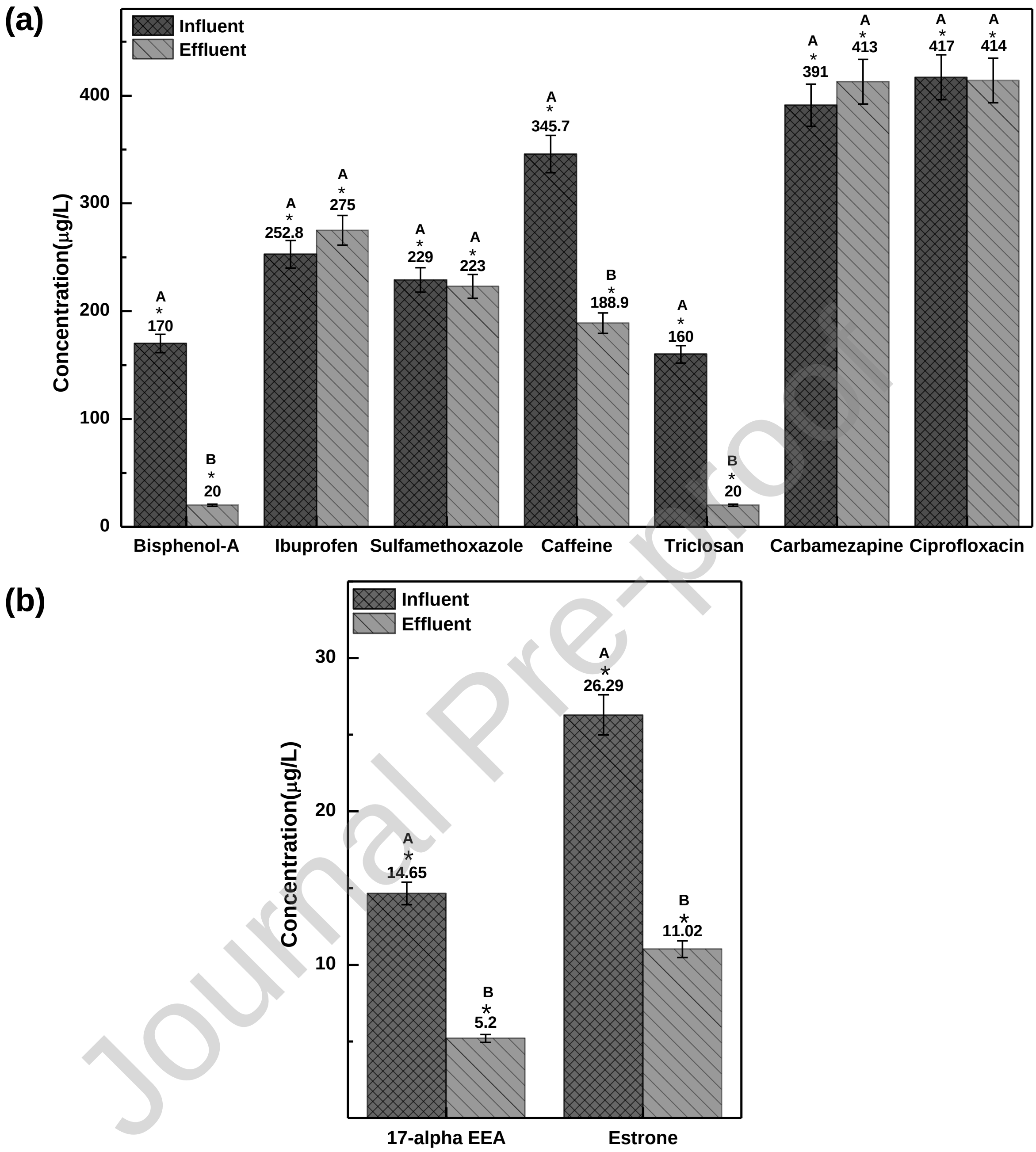
(a)
Influent
Effluent
Concentration(μg/L)
0
100
200
300
400
A
*
170
B
*
20
A
*
252.8
A
*
275
A
*
229
A
*
223
A
*
345.7
B
*
188.9
A
*
160
B
*
20
A
*
391
A
*
413
A
*
417
A
*
414
Bisphenol-A
Ibuprofen
Sulfamethoxazole
Caffeine
Triclosan
Carbamezapine
Ciprofloxacin
(b)
Influent
Effluent
Concentration(μg/L)
10
20
30
A
*
14.65
B
*
5.2
A
*
26.29
B
*
11.02
17-alpha EEA
Estrone

Figure 3

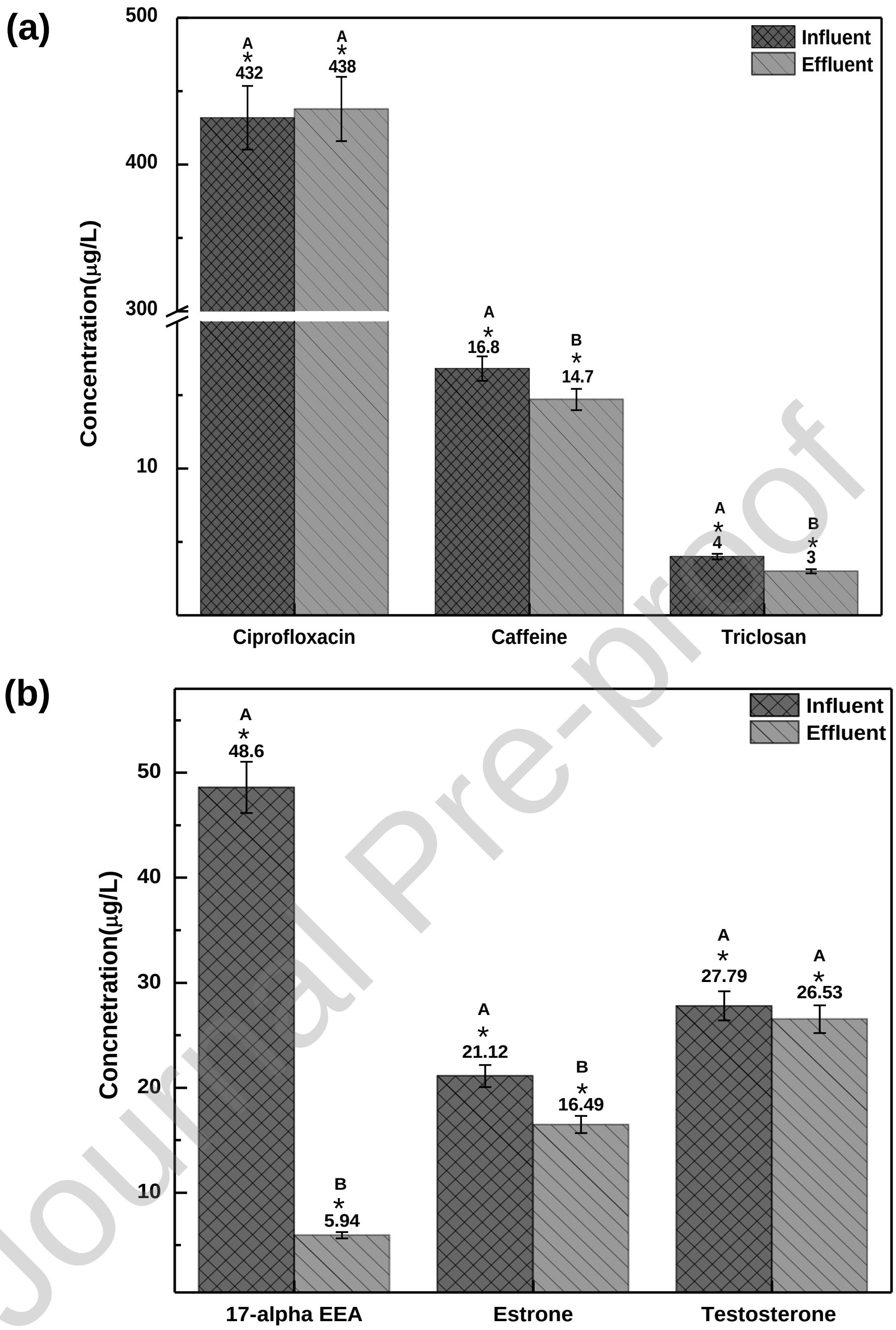
(a)
500
400
300
10
Concentration(μg/L)
Influent
Effluent
A
*
432
A
*
438
A
*
16.8
B
*
14.7
A
*
4
B
*
3
Ciprofloxacin
Caffeine
Triclosan
(b)
50
40
30
20
10
Concnetration(μg/L)
A
*
48.6
B
*
5.94
A
*
21.12
B
*
16.49
A
*
27.79
A
*
26.53
17-alpha EEA
Estrone
Testosterone

Figure 4

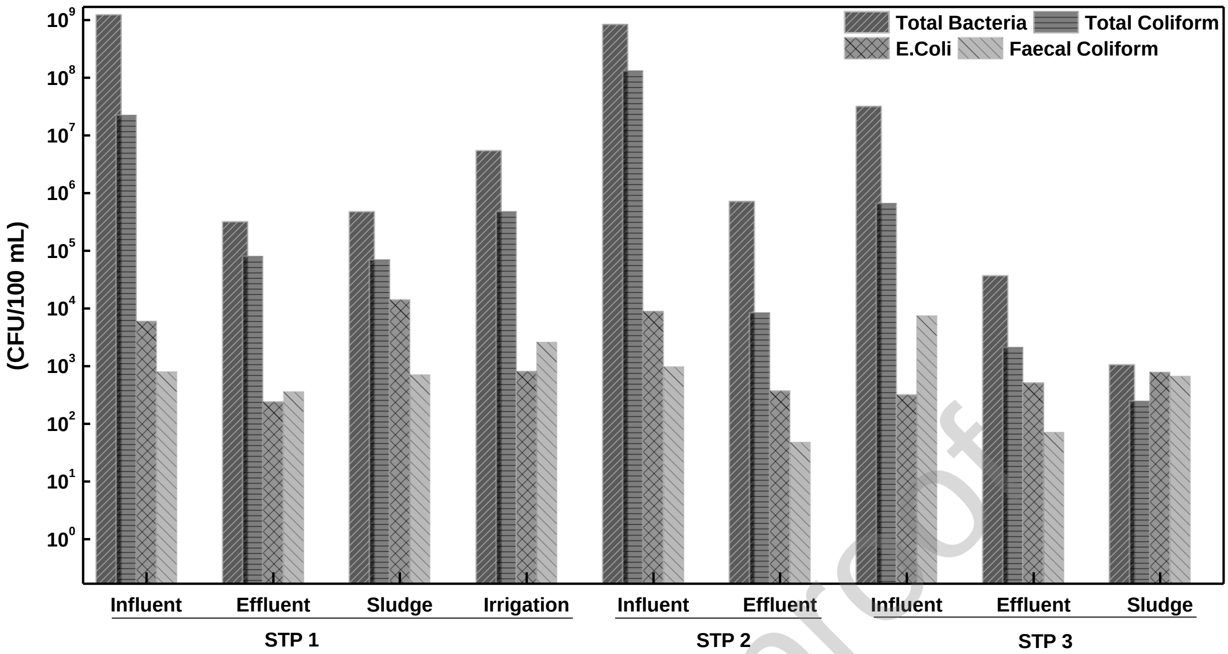

Figure 5

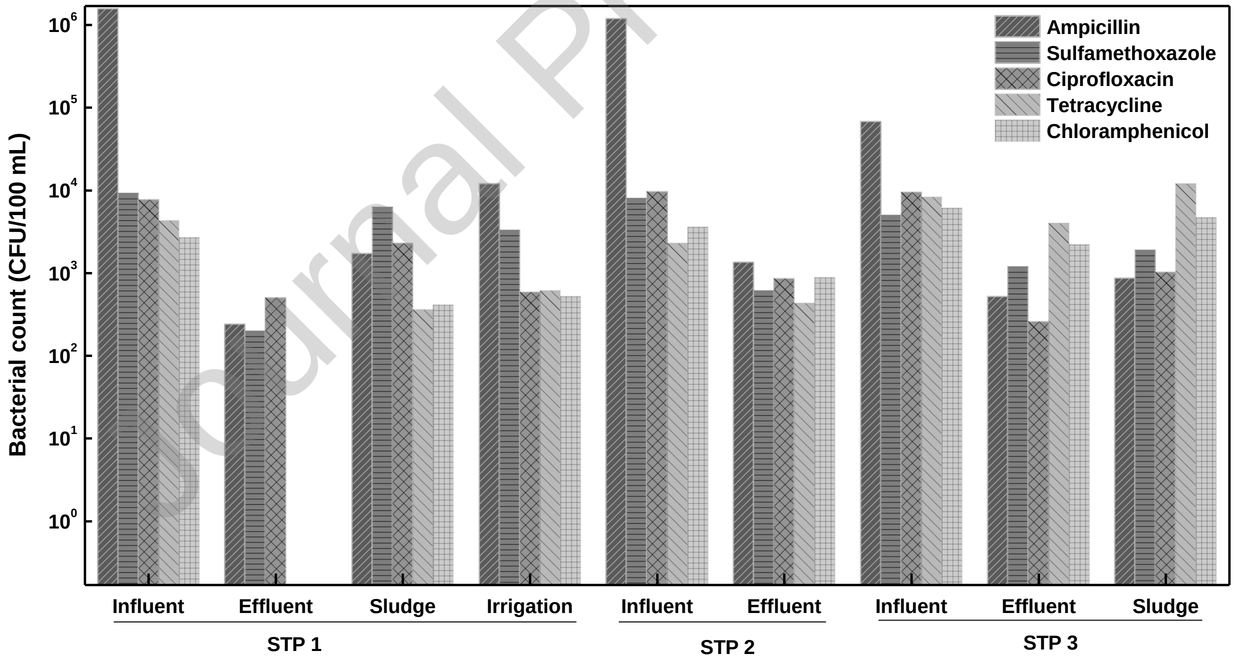

Figure 6

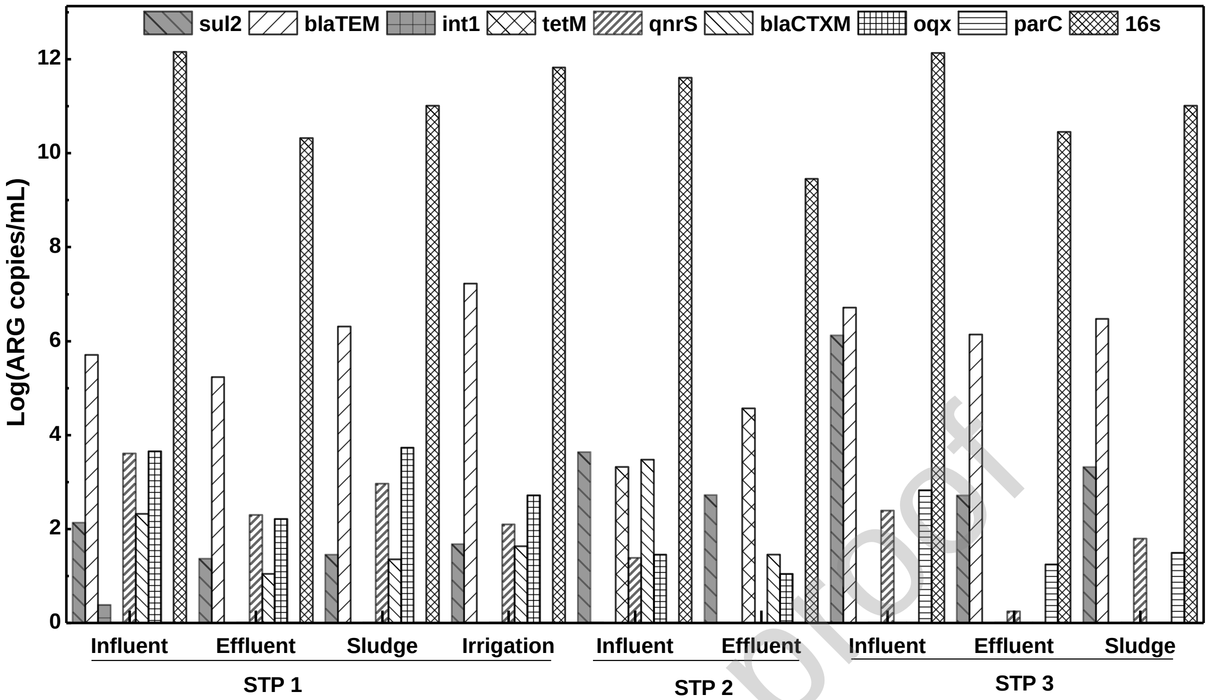

Figure 7

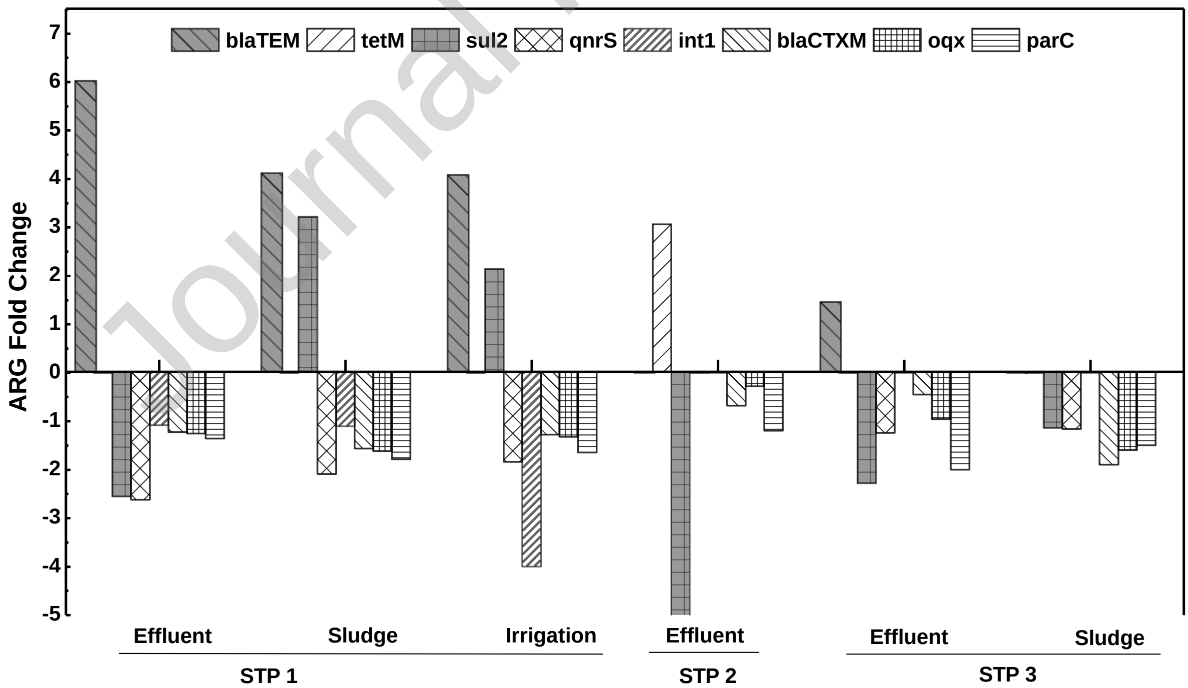

Figure 8

Table 1: Targeted CECs, Molecular Weights and Limit of Detection & Quantification

Analytes	Molecular weight (g/mol)	Retention Time (min)	LOD (µg/L)	LOQ (µg/L)
Caffeine	194.1	1.9	42.8	129.8
Carbamazepine	236.2	2.4	5.7	17.2
Ciprofloxacin	331.3	2.7	30.0	100.0
Bisphenol-A	228.2	2.9	3.1	10.5
Estrone	270.3	4.0	0.3	0.9
Ibuprofen	206.2	1.8	166.1	503.4
Sulphamethoxazole	253.2	1.7	52.0	156.0
Triclosan	290.3	1.8	15.0	45.0
17 α - ethynylestradiol	296.4	3.8	0.2	0.6
Testosterone	288.4	3.3	74.0	246.0
Progesterone	314.4	4.2	2.0	5.0

CREDIT authorship contribution statement

- Priyam Saxena: Investigation, Methodology, Writing - review & editing.
- Isha Hiwrale: Investigation, Methodology, Writing - review & editing.
- Sanchita Das: Investigation, Methodology, Writing - review & editing.
- Varun Shukla: Investigation, Methodology.
- Lakshay Tyagi: Investigation, Writing - review & editing
- Nishant Dafale: Supervision, Visualization, Conceptualization, Writing - review & editing, Writing - original draft.
- Sukdeb Pal: Supervision, Visualization, Conceptualization, Writing - review & editing.
- Rita Dhodapkar: Supervision, Conceptualization, Funding acquisition, Writing - original draft

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Highlights

- Contamination profiles of 3 STPs in urban and semi-urban areas in India.
- PPCPs detected in treated and untreated sewage over a wide range of concentrations.
- Antibiotic resistance bacteria and genes detected in treated wastewater.